CS & IT ENGINEERING



Algorithms

Analysis of Algorithms



Lecture No.- 08

By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Properties

Practice Questions

Topics to be Covered



- Topic
- Topic
- Topic

Practice Questions

Algo Analysis



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Telegram



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

[MSQ]

#Q. Asymptotic Comparison of 2 functions:

$f(n) = n$

$g(n) = n \log n$

$n < n \log n$

A $f = O(g)$

B $f = o(g)$

C $f = \Omega(g)$ X

D $f = \omega(g)$ X

A, B

[MCQ]

#Q. Arrange the given functions in increasing order of rate of growth:

$f_1 \to 2^n$, $f_2 \to n^{3/2}$, $f_3 \to n \log n$, $f_4 \to n^{(\log n)}$

A f_2, f_3, f_1, f_4

B f_3, f_2, f_1, f_4

C f_3, f_2, f_4, f_1

D f_2, f_3, f_4, f_1

$n\sqrt{n}$ $\quad f_2 > f_3$

$n \log n$ $\quad f_1 > f_4$

$n^{\log n}$

2^n $n^{\log n}$

$n > (\log n)^2$

[NAT]

#Q. You are given a database having 10^x records.

There are 2 packages available for processing the data.

Package A takes a time of 10 * n * log n while,

Package B takes a time of 0.0001 * n^2 for processing 'n' records.

Determine the smallest integer x for which package A outperforms package B.

24%

Solve NAT

A better than B

$t_A < t_B$

A -

B -

C -

D -

Ans - 6

$t_A = 10 \times n \times \log_{10} n$

$t_B = 0.0001 * n^2$

$n = 10^x$

$x = 1$

$n = 10^1 = 10$

$t_A > t_B$

$t_A = 10 \times 10 \log_{10} 10 = 100$

$t_B = 10^{-4} * n^2 = 10^{-4} \times (10)^2 = 10^{-2} = 0.01$

6×10^7 $\quad$ 10^8

6×10^7 $\quad$ 10×10^7

$6 < 10$

$x = 6$

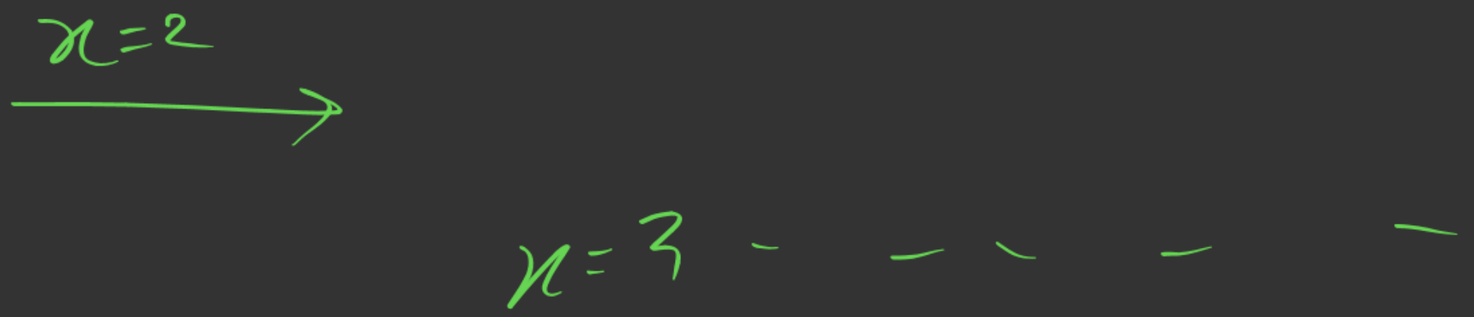
$t_A = 10 \times 10^6 \times \log_{10} 10^6$

$= 10^7 \times 6$

$t_B = 10^{-4} \times (10^6)^2$

$= 10^{-4} \times 10^{12}$

$= 10^{12-4} = 10^8$


n=2
n=3

$t_A < t_B$

$10 \times n \times \log_{10} n < 10^{-4} * (n)^2$

$n = 10^x$

Useful →

$10 \times 10^x \times \log_{10}(10^x) < 10^{-4} \times (10^x)^2$

$x \times 10^{x+1} < 10^{(2x-4)}$

$x < 10^{(2x-4-x-1)}$

$x < 10^{(x-5)}$

$n = 6$

$x = 5$ ✗

$5 < 10^{(5-5)}$

$5 < 10^0$

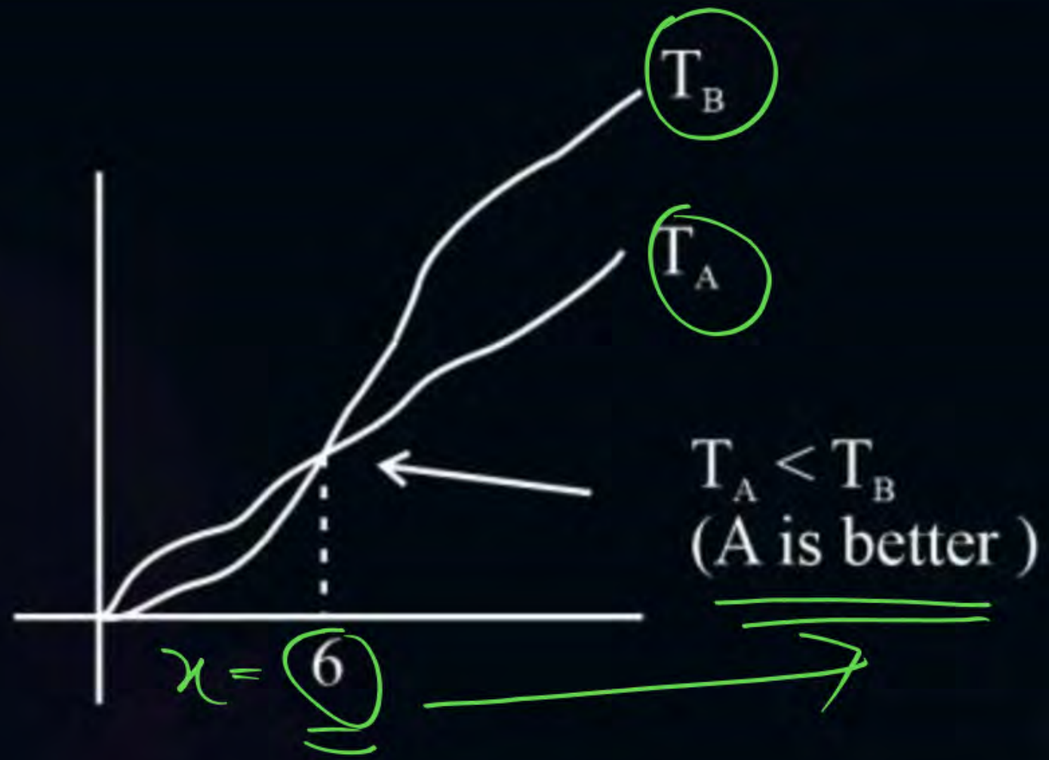
$5 < 1$

$x = 6$

$6 < 10^{6-5}$

$6 < 10$

True


T_B
T_A
$T_A < T_B$
(A is better)
$x = 6$

{ Analysis of an Algo whose Problem Statement is given }

```
Algo LinearSearch(A, n, x)
{
    for(i=0; i<n; i++)
    {
        if(A(i)==x)
            return i
    }
    return -1;
}
```

Analyse

Best Case
Worst Case

$\Omega(1)$
$O(n)$

$O(1)$

B.C

$n=4$

A : | 5 | 2 | 3 | 9 |

$x=5$

1

W.C

A : | 5 | 2 | 3 | 9 |

$n=9$, $x=1$

$O(n)$

[NAT]

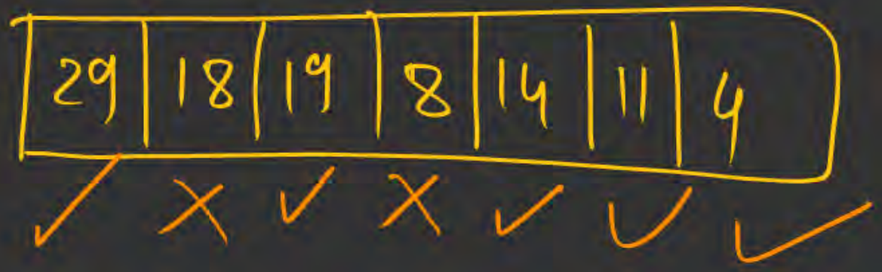
#Q. An element in an Array is called Leader if it is greater than all elements to the right of it. The time complexity of the most efficient algorithm to print all Leaders of the given Array of size 'n' is _____.

V. V. Imp

(Interview)

⭐ Given: last elem is always a leader

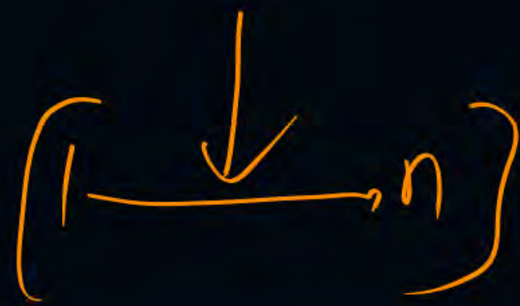
edge case

eg- A = 

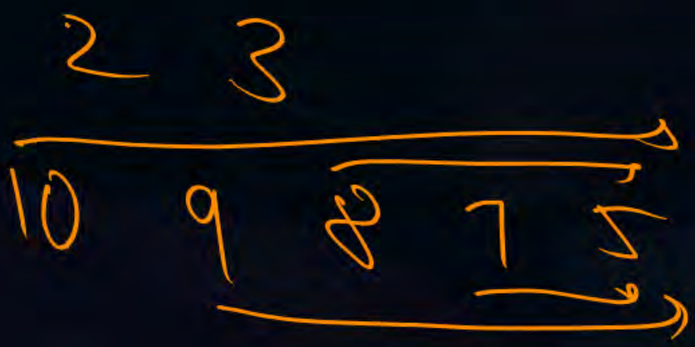
[29, 19, 14, 11, 4]

Solution 1: Brute Force

```
Algo AJLeader(A[], n) {
    for (i = 1; i < n; i++){
        for (j = i+1; j ≤ n; j++) {
            if (A[i] < A[j]) {
                break
            }
        }
        if (j == (n+1)) {
            print(A[i])
        }
    }
}
```



1 based indexing





Topic : Analysis of Algorithms

Time Complexity Analysis of Brute Force Approach:

(1) Best Case:

A is in increasing order.

E.g.: A = 5, 10, 12, 18

Total Comparisons → (n – 1) * O(1) = O(n – 1)

→ O(n)



Topic : Analysis of Algorithms

Time Complexity Analysis of Brute Force Approach:

(2) Worst Case:

A is in decreasing order.

E.g.: A = 18, 15, 12, 10, 5

Total Comparisons → 4 + 3 + 2 + 1

In general → $(n-1) + (n-2) + (n-3) \ldots 1 = \frac{n(n+1)}{2} - n$

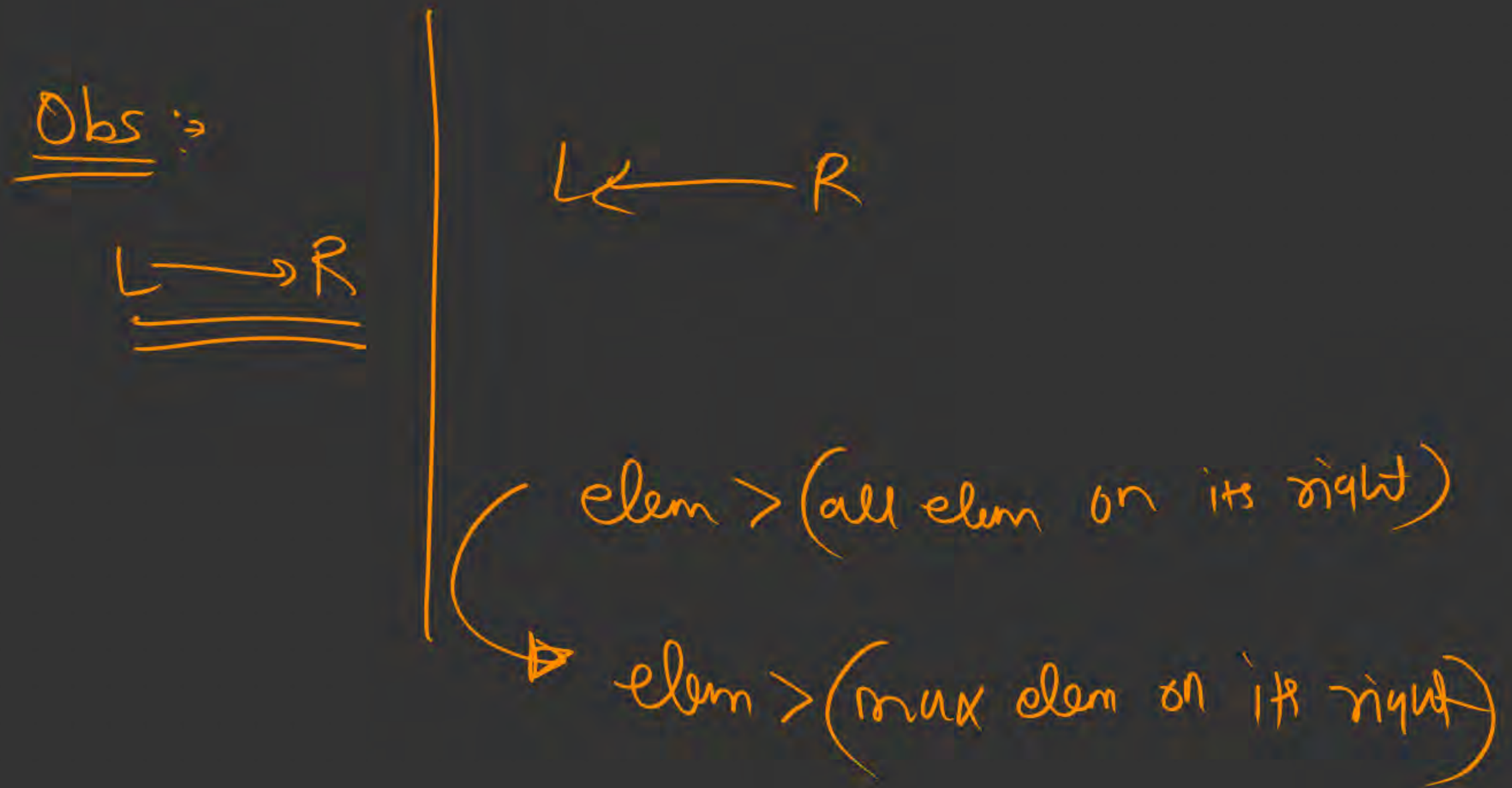
$$= \frac{n^2 - n}{2} \rightarrow O(n^2)$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Brute Force

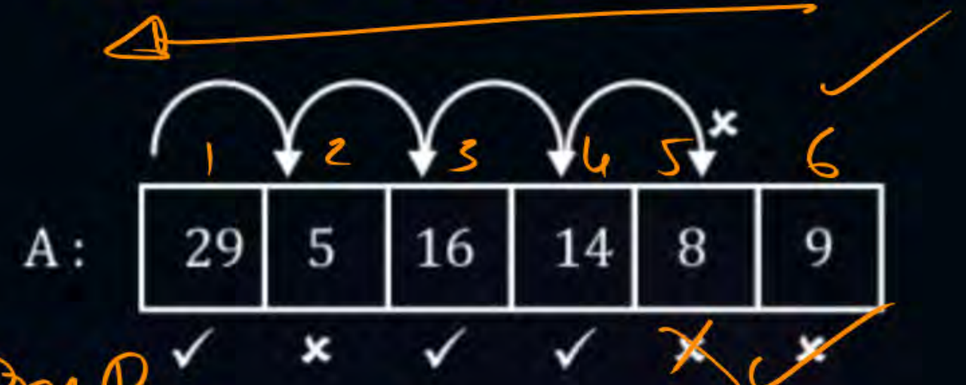
$B.C = \Omega(n)$

$W.C = O(n^2)$


Obs :
L ⟶ R
L ⟵ R
elem > (all elem on its right)
elem > (max elem on its right)

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Code walk through



Dry Run: L = 9

8 > 9 → ✗

14 > 9 → ✓ → L = 14

16 > 14 → ✓ → L = 16

5 > 16 → ✗

29 > 16 → L = 29

Algo. AJ Leader (a, n)

```
{
    print(A[n])
    L= A[n]        → O(1)
    for (I = (n-1); I >=1; I--)
    {
        if(A [i] > L)
        {
            L= A[i]
            print (A[i])
        }
    }
}
```

O(n)



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Time complexity Analysis

1. Best Case:- Increasing order → $O(n) \rightarrow \Omega(n)$
2. Worst Case:- Decreasing order → $O(n) \rightarrow O(n)$

} $O(n)$

Imp. Irrespective of the input order the Algo will take same amount fine time → $O(n)$

TC ⇒ efficient [redacted] algo $O(n)$

9 PM

1 hr 1000 - 3 hrs



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Summary

	Best Case	Worst Case
1. Brute force	$O(n)$ → $\Omega(n)$	$O(n^2)$
2. Efficient Algo	$O(n)$ → $\Omega(n)$	$O(n)$

(Prev) (Api)

$\Theta(n)$

Best WC

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#Q. Arrange the following function in increasing order of their growth rate?

$f_1 \to nlogn,\ f_2 \to n^2,\ f_3 \to e^n,\ f_4 \to n^{3/2},\ f_5 \to n,\ f_6 \to (1/n),\ f_7 \to 2^n$

Poly P E P P D E

$n^2 > n^{3/2}$

$n < nlogn$

$f_3 > f_7$

$e^n \quad 2^n$

$2.71 > 2$

A. $F_2, F_1, F_4, F_3, F_5, F_7, F_6$

B. $F_6, F_5, F_1, F_4, F_2, F_3, F_7$

C. $F_6, F_5, F_1, F_4, F_2, F_7, F_3$ ✓

D. $F_6, F_5, F_1, F_4, F_7, F_3, F_2$



Summary

Imp Concepts



THANK - YOU